# Discovery

## (TA0007)

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# Discovery

- The adversary is trying to figure out your environment.

# Discovery

- Discovery consists of techniques an adversary may use to gain knowledge about the system and internal network.
- These techniques help adversaries observe the environment and orient themselves before deciding how to act.
- They also allow adversaries to explore what they can control and what's around their entry point in order to discover how it could benefit their current objective.
- Native operating system tools are often used toward this post-compromise information-gathering objective.

# Discovery: Techniques

- Account Discovery
- Application Window Discovery
- Browser Information Discovery
- Cloud Infrastructure Discovery
- Cloud Service Dashboard
- Cloud Service Discovery
- Cloud Storage Object Discovery
- Container and Resource Discovery
- Debugger Evasion

# Discovery: Techniques

- Device Driver Discovery
- Domain Trust Discovery
- File and Directory Discovery
- Group Policy Discovery
- Log Enumeration
- Network Service Discovery
- Network Share Discovery
- Network Sniffing
- Password Policy Discovery

# Discovery: Techniques

- Peripheral Device Discovery
- Process Discovery
- Query Registry
- Remote System Discovery
- Software Discovery
- System Information Discovery
- System Location Discovery
- System Network Configuration Discovery
- System Network Connections Discovery

# Discovery: Techniques

- System Owner/User Discovery
- System Service Discovery
- System Time Discovery
- Virtualization/Sandbox Evasion

# Discovery: Account Discovery

- Adversaries may attempt to get a listing of valid accounts, usernames, or email addresses on a system or within a compromised environment.
- This information can help adversaries determine which accounts exist, which can aid in follow-on behavior such as brute-forcing, spear-phishing attacks, or account takeovers (e.g., Valid Accounts).

# Discovery: Account Discovery

- Adversaries may use several methods to enumerate accounts, including abuse of existing tools, built-in commands, and potential misconfigurations that leak account names and roles or permissions in the targeted environment.

# Discovery: Account Discovery

- For examples, cloud environments typically provide easily accessible interfaces to obtain user lists.
- On hosts, adversaries can use default PowerShell and other command line functionality to identify accounts.
- Information about email addresses and accounts may also be extracted by searching an infected system’s files.

# Discovery: Account Discovery: Sub-techniques

- Local Account
- Domain Account
- Email Account
- Cloud Account

# Discovery: Account Discovery: Local Account

- Adversaries may attempt to get a listing of local system accounts.
- This information can help adversaries determine which local accounts exist on a system to aid in follow-on behavior.
- Commands such as net user and net localgroup of the Net utility and id and groups on macOS and Linux can list local users and groups.
- On Linux, local users can also be enumerated through the use of the /etc/passwd file.
- On macOS the dscl . list /Users command can be used to enumerate local accounts.

# Discovery: Account Discovery: Local Account 실습

- 윈도우(파워셸)
    - net user
    - net groups
- 리눅스
    - cat /etc/passwd
    - cat /etc/group

- Procedure Examples
    - https://attack.mitre.org/techniques/T1087/001/

# Discovery:  Remote System Discovery

- Adversaries may attempt to get a listing of other systems by IP address, hostname, or other logical identifier on a network that may be used for Lateral Movement from the current system.
- Functionality could exist within remote access tools to enable this, but utilities available on the operating system could also be used such as Ping or net view using Net.

# Discovery:  Remote System Discovery

- Adversaries may also analyze data from local host files (ex: C:\Windows\System32\Drivers\etc\hosts or /etc/hosts) or other passive means (such as local Arp cache entries) in order to discover the presence of remote systems in an environment.

# Discovery:  Remote System Discovery

- Adversaries may also target discovery of network infrastructure as well as leverage Network Device CLI commands on network devices to gather detailed information about systems within a network (e.g. show cdp neighbors, show arp).

# Discovery: Remote System Discovery 실습

- 공통
  - ping
  - arp
- 윈도우
  - notepad C:\Windows\System32\Drivers\etc\hosts
- 리눅스
  - cat /etc/hosts

- Procedure Examples
  - https://attack.mitre.org/techniques/T1018/

# Discovery: Network Service Discovery

- Adversaries may attempt to get a listing of services running on remote hosts and local network infrastructure devices, including those that may be vulnerable to remote software exploitation.
- Common methods to acquire this information include port and/or vulnerability scans using tools that are brought onto a system.

# Discovery: Network Service Discovery

- Within cloud environments, adversaries may attempt to discover services running on other cloud hosts.
- Additionally, if the cloud environment is connected to a on-premises environment, adversaries may be able to identify services running on non-cloud systems as well.

# Discovery: Network Service Discovery

- Within macOS environments, adversaries may use the native Bonjour application to discover services running on other macOS hosts within a network.
- The Bonjour mDNSResponder daemon automatically registers and advertises a host’s registered services on the network.
- For example, adversaries can use a mDNS query (such as dns-sd -B _ssh._tcp .) to find other systems broadcasting the ssh service

# Discovery: Network Service Discovery 실습

- 네트워크/포트 스캔 도구
  - nmap

- Procedure Examples
  - https://attack.mitre.org/techniques/T1046/